

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AC1-003***

Tidd-Wylie Ridge 345 kV

April 2018

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

ESC Brooke County Power I, LLC (“Interconnection Customer”) is requesting 80 MW incremental increase (uprate) to prior queue requests AA2-121 and AB2-104. The power plant is a 2x1 combined cycle natural gas generation facility located at approximately 10.4 miles south of Wylie Ridge substation on the Tidd-Wylie Ridge 345 kV line in Brooke County, West Virginia. The increased capability associated with queue position AC1-003 is achieved through improved steam turbine (ST Generator) duct firing application that maximizes the utilization of existing equipment and increases facility output.

The following table shows the MW unit contributions as related to the Maximum Facility Output:

	Queue No.	CT1 Unit	CT2 Unit	ST Unit	Total MW
	AA2-121/AB2-104	246	246	258	750
	AC1-003	19	19	42	80
Total MFO MW		265	265	300	830

The following table shows the Capacity Interconnection Rights MWs by unit:

Queue No.	CT Unit1	CT Unit2	ST Unit	Total CIRs
AA2-121	224.1 MW CIRs, commencing June 1, 2019	224.1 MW CIRs, commencing June 1, 2019	236.8 MW CIRs, commencing June 1, 2019	685 MW CIRs, commencing June 1, 2019
AB2-104	21.9 MW CIRs, commencing June 1, 2020	21.9 MW CIRs, commencing June 1, 2020	21.2 MW CIRs, commencing June 1, 2020	65 MW CIRs, commencing June 1, 2020
AC1-003	19 MW CIRs, commencing June 1, 2020	19 MW CIRs, commencing June 1, 2020	19 MW CIRs, commencing June 1, 2020	80 MW CIRs, commencing June 1, 2020
Total CIR MW	Combined 265 MW CIRs as of June 1, 2020	Combined 265 MW CIRs as of June 1, 2020	Combined 300 MW CIRs as of June 1, 2020	Combined 830 MW CIRs as of June 1, 2020

The proposed in-service date for the AC1-003 (and AA2-121/AB2-104) projects is 6-1-2020. **This study does not imply a Monongahela Power Company (“Transmission Owner” or “Mon Power”) to this in-service date.**

For convenience, the following table references the published study reports of associated queue requests AA2-121 and AB2-104

Queue No.	Type of Study Report	Website Link at PJM Website
AA2-121	Facilities Study Report	http://www.pjm.com/pub/planning/project-queues/facilities/aa2121_fac.pdf
AB2-104	System Impact Study Report	http://www.pjm.com/pub/planning/project-queues/impact_studies/ab2104_imp.pdf

Point of Interconnection (POI)

AC1-003 (and AA2-121/AB2-104) will interconnect with the Mon Power transmission system by tapping the Tidd-Wylie Ridge 345 kV line at a point located approximately 10.4 miles from Wylie Ridge substation. A new three breaker ring bus station, named Abby Lane, will be built for this interconnection. Please refer to Appendix 2 for a simplified interconnection single line diagram.

Transmission Owner Scope of Work and Costs Summary

Mon Power facilities and network upgrades required to support AC1-003 project are as follows:

- (a) Attachment Facilities: \$0.00
- (b) Direct Connection Network Upgrades: \$0.00
- (c) Non-Direct Connection Network Upgrades: \$0.00*
 - (c1) Wylie Ridge 345 kV Substation. Adjust remote relay and metering settings at the Wylie Ridge 345 kV substation.
PJM Network Upgrade No. N5255.....\$6,100
 - (c2) Abby Lane 345 kV Substation. Adjust remote relay and metering settings at the Abby Lane 345 kV substation.
PJM Network Upgrade No. N5256.....\$6,100
- (d) Direct Connection Local Upgrades: \$0.00
- (e) Non-Direct Connection Local Upgrades: \$0.00
- (f) Option to Build Upgrades: \$0.00

Estimated Total Costs (a) to (f):\$12,200

NOTE 1: The costs listed above (\$12,200) were paid by prior queue request AB2-104. For more details, please refer to the AB2-104 system impact study report which can be found at:

http://www.pjm.com/pub/planning/project-queues/impact_studies/ab2104_imp.pdf

NOTE 2: Above amounts are net and in 2017 Dollars. The Federal Income Tax Gross Up charge may or may not be added to above amounts depending on whether this project meets the eligibility requirements of IRS Safe Harbor Provisions for Non-Taxable Status.

Interconnection Customer Requirements

In addition to construction work listed in the system impact report and facilities study report of prior queue requests, AB2-104 and AA2-121, Interconnection Customer will also be responsible for meeting all criteria as specified in the applicable sections of FirstEnergy "Requirements for Transmission Connected Facilities" document which can be found at PJM's website at <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx> , including:

1. A compliance with the FirstEnergy and PJM generator power factor and voltage control requirements. The analysis of the estimated reactive capability curves shows that the increase in capacity for the AC1-003 request results in a dynamic reactive deficiency for the ST machine. For the AC1-003 request, the power factor for the increased output must meet 1.0 (unity) to 0.90 lagging requirements measured at the generator terminals.
2. The execution of a back-up service agreement to serve the customer load supplied from the AC1-003 generation project metering point when the units are out-of-service. This assumes the intent of Interconnection Customer is to net the generation with the load.

The above requirements are in addition to any metering, any requirements for the AA2-121/AB2-104 Generation Projects, or any other requirements imposed by PJM.

Revenue Metering and SCADA Requirements

PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

FirstEnergy Requirements

AC1-003 will be required to comply with all FirstEnergy revenue metering requirements for generation interconnection customers. The FE revenue metering requirements may be found in the FirstEnergy "Requirements for Transmission Connected Facilities" document located at the following link:

<http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>

Because the AC1-003 generation project connection is completely on the customer's side of the AA2-121/AB2-104 project's POI, it is assumed no new metering requirements will be necessary beyond the requirements in the AA2-121/AB2-104 Interconnection Service Agreement ("ISA").

Schedule

Based on the scope of the attachment facilities, direct and non-direct system upgrades, it is expected to take a minimum of eight (8) months from the date of a fully executed Interconnection Construction Service Agreement to complete the installation. This includes a full payment for the work that is related to this interconnection project. It also assumes that the Interconnection Customer will provide the property for the Project direct connection facilities and all right-of-way, permits, easements, etc. that will be needed. A further assumption is that there will be no environmental issues with any of the new properties associated with this project, that there will be no delays in acquiring the necessary permits for implementing the defined direct connection facilities and that transmission system outages will be possible when requested.

Network Impacts

The Queue Project AC1-003 was evaluated as a 80.0 MW (Capacity 80.0 MW) injection into the AA2-121 345 kV substation (tap of the Tidd – Wylie Ridge 345 kV line) in the APS area. Project AC1-003 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AC1-003 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Analysis - 2020

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Steady-State Voltage Requirements

None

Short Circuit

None

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request. Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

Not Applicable

Light Load Analysis - 2020

Not Required.

System Reinforcements

Short Circuit

None

Stability and Reactive Power Requirement

Mitigation is not required. Please refer to Appendix 3 for Dynamic Simulation Analysis Report.

Summer Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

None

Light Load Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

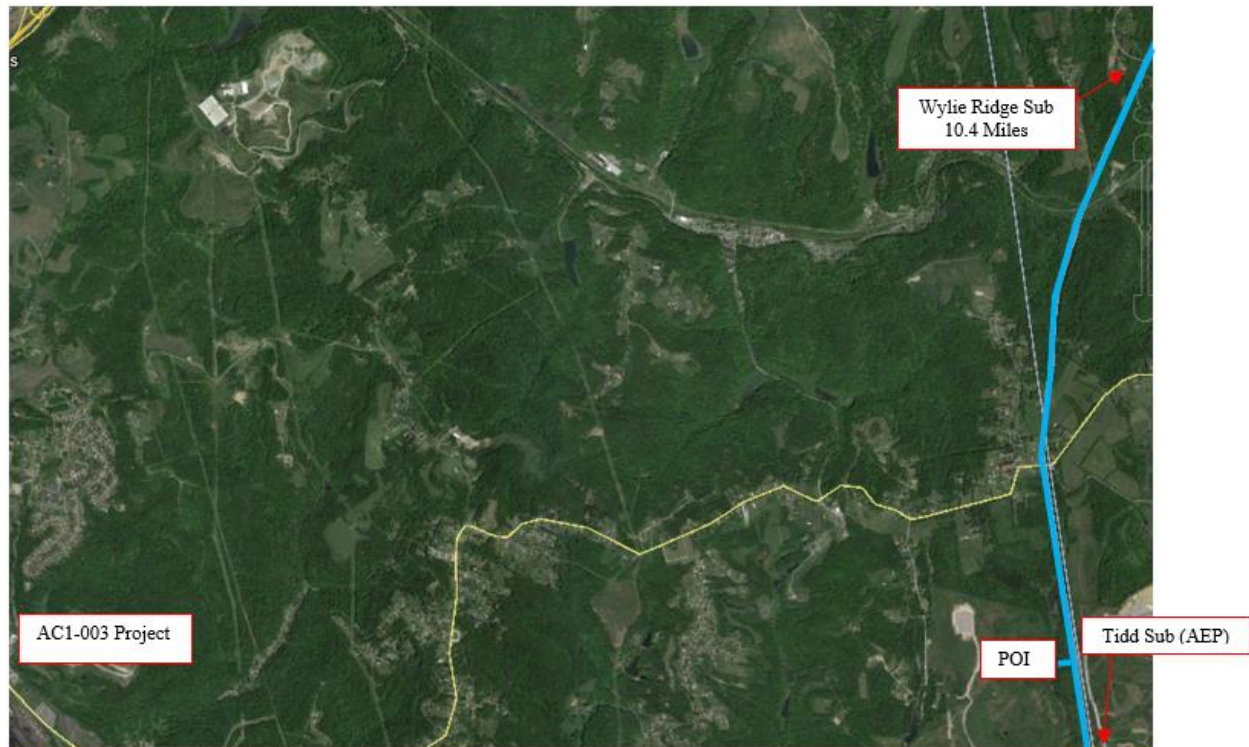
(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

None

Appendix 1

Facility Location

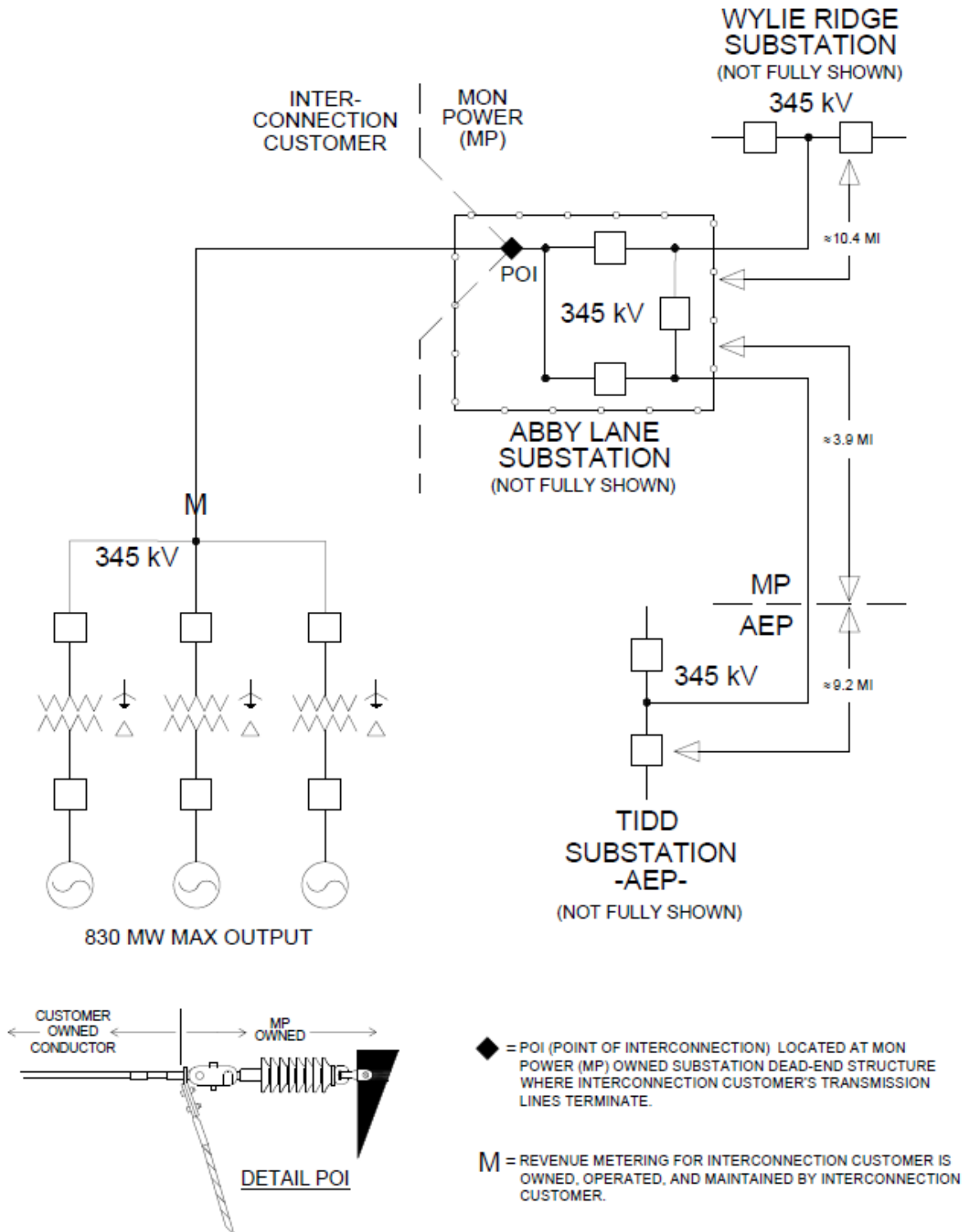
PJM Queue Position: AC1-003



Appendix 2

Interconnection One-Line Diagram

PJM Queue Position: AC1-003



Appendix 3

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AC1-003

Executive Summary

Generator Interconnection Request AC1-003 is for an increase in energy injection capability of the 2x1 combined cycle facility interconnection request AB2-104. The uprate increases the Maximum Facility Output (MFO) of the plant from 750 MW to 830 MW. AC1-003 has a Point of Interconnection (POI) at the existing AB2-104 345 kV Substation, which taps the 345 kV Tidd – Wylie Ridge circuit in the Allegheny Power System (APS) (part of First Energy), in Brooke County, West Virginia.

This report describes a dynamic simulation analysis of AC1-003 as part of the overall system impact study.

The load flow scenario for the analysis is based on the RTEP 2020 Light Load case, modified to include applicable queue projects. AC1-003 was set to maximum power output.

AC1-003 was tested for compliance with NERC, PJM, Transmission Owner and other applicable criteria. 90 contingencies were studied, each with at least a 20 second simulation time period.

Studied scenarios included:

- a) Steady state operation;
- b) Three phase faults with normal clearing time;
- c) Single phase faults with stuck breaker;
- d) Single-phase faults placed at 80% of the line with delayed (Zone 2) clearing at line end remote from the fault due to primary communications/relay failure.
- e) Single-phase faults with loss of multi-circuit tower line.
- f) Three phase faults with high speed reclosing (HSR)

No relevant bus contingencies were found.

For all simulations, the queue project under study along with the rest of the PJM system were required to maintain synchronism and with all states returning to an acceptable new condition following the disturbance.

For the fault contingencies tested on the 2020 Light Load case:

- a) Post-contingency oscillations were insufficiently damped for all contingencies. Pre- and post- AC1-003 damping analysis carried out on the CDG3 unit (bus 243185) showed that the oscillations remained, therefore this is not attributable to the addition of AC1-003.
- b) The AC1-003 generator was able to ride through all faults (except for faults where protective action trips a generator(s)).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of the fault.

No mitigations were found to be required.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AC2-142

1. Introduction

Generator Interconnection Request AC1-003 is for an increase in energy injection capability of the 2x1 combined cycle facility interconnection request AB2-104. The uprate increases the Maximum Facility Output (MFO) of the plant from 750 MW to 830 MW. AC1-003 has a Point of Interconnection (POI) at the existing AB2-104 345 kV Substation, which taps the 345 kV Tidd – Wylie Ridge circuit in the Allegheny Power System (APS) (part of First Energy), in Brooke County, West Virginia.

This analysis is effectively a screening study to determine whether the addition of AC1-003 will meet the dynamics requirements of the NERC, PJM and Transmission Owner reliability standards.

In this report, the AC1-003 project and how it is proposed to be connected to the grid are first described, followed by a description of how the project is modeled in this study. The fault cases are then described and analyzed, and lastly a discussion of the results is provided.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AC2-142

2. Description of the Project

Generator Interconnection Request AC1-003 is for an increase in energy injection capability of the 2x1 combined cycle facility interconnection request AB2-104. The uprate increases the Maximum Facility Output (MFO) of the plant from 750 MW to 830 MW.

AC1-003 has a Point of Interconnection (POI) at the existing AB2-104 345 kV Substation, which taps the 345 kV Tidd – Wylie Ridge circuit in the Allegheny Power System (APS) (part of First Energy), in Brooke County, West Virginia.

Figure 1 shows the simplified single-line diagram of the AC1-003 loadflow model.

Table 1 lists the parameters given in the impact study data and the corresponding parameters of the AC1-003 loadflow model.

Additional project details are provided in Attachments 1 through 4:

- Attachment 1 contains the Impact Study Data which details the proposed AC1-003 project.
- Attachment 2 shows the one-line diagram of the APS system in the vicinity of AC1-003.
- Attachment 3 provides a diagram of the PSS/E model in the vicinity of AC1-003.
- Attachment 4 gives the PSS/E loadflow and dynamic models of the AC1-003 plant.

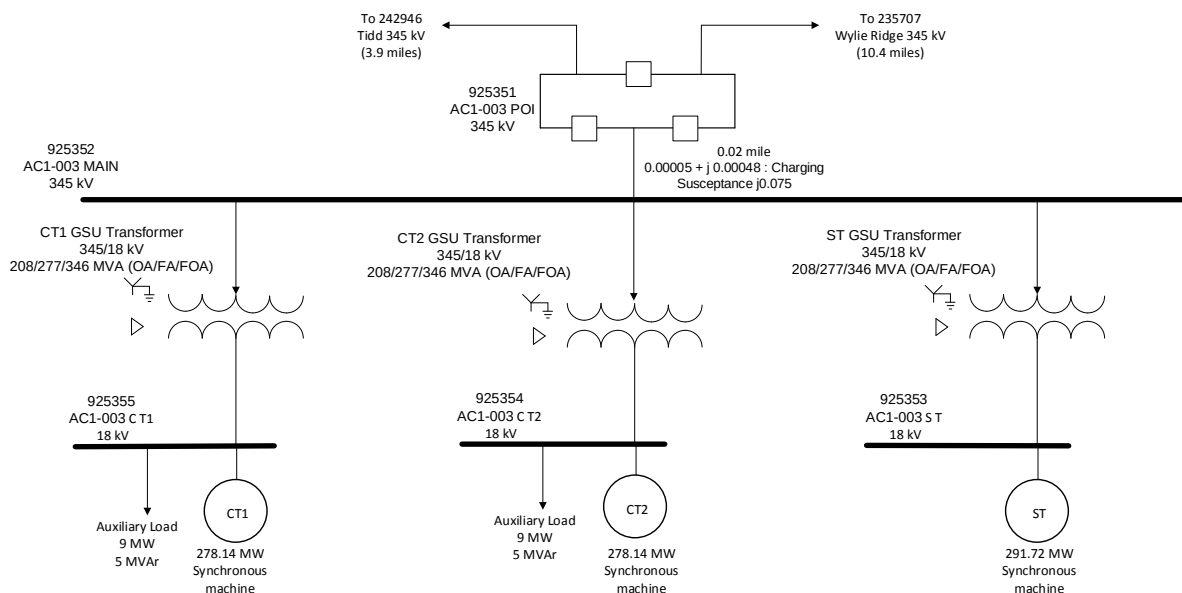


Figure 1: AC1-003 Plant Model

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AC2-142

Table 1: AC1-003 Plant Model

Element	Impact Study Data	Model
Combustion turbines – CT1, CT2	2 x 278.14 MW combustion turbine generators MVA base = 330 MVA Vt = 18 kV Unsaturated sub-transient reactance = 0.215 pu @ MVA base	2 x 278.14 MW Generators Pgen 278.14 MW Pmax 278.14 MW Pmin 0 MW Qgen 45.472 MVar Qmax 175 MVar Qmin -120 MVar Mbase 330 MVA Zsource 0.0028 + j0.215 pu @ Mbase
Steam turbine ST	291.72 MW steam turbine generator MVA base = 330 MVA Vt = 18 kV Unsaturated sub-transient reactance = 0.215 pu @ MVA base	291.72 MW Generator Pgen 291.72 MW Pmax 291.72 MW Pmin 0 MW Qgen 45.472 MVar Qmax 155 MVar Qmin -115 MVar Mbase 330 MVA Zsource 0.0028 + j0.215 pu @ Mbase
Combustion turbine generator GSU transformers	2 x 345/18 kV transformers Rating = 208/277/346 MVA (ONAN/ONAF/ONAF) Transformer base = 208 MVA Impedance = 0.002 + j0.100 pu @ MVA base Number of taps = N/A Tap step size = N/A	2 x 345/18 kV transformers Rating = 208/277/346 MVA (ONAN/ONAF/ONAF) Transformer base = 208 MVA Impedance = 0.002 + j0.100 pu @ MVA base Number of taps = 5 Tap step size = 2.5 %
Steam turbine generator GSU transformer	345/18 kV transformer Rating = 208/277/346 MVA (ONAN/ONAF/ONAF) Transformer base = 208 MVA Impedance = 0.002 + j0.100 pu @ MVA base Number of taps = N/A Tap step size = N/A	345/18 kV transformer Rating = 208/277/346 MVA (ONAN/ONAF/ONAF) Transformer base = 208 MVA Impedance = 0.002 + j0.100 pu @ MVA base Number of taps = 5 Tap step size = 2.5 %
Auxiliary load	9 MW + 5 MVar at each CT	9 MW + 5 MVar at each CT 1
Transmission line	0.02 Miles 0.00005 + j 0.00048 Charging Susceptance: 0.075 j all impedance @ 100 MVA base	0.02 Miles 0.00005 + j 0.00048 Charging Susceptance: 0.075 j all impedance @ 100 MVA base ²

¹ Auxiliary loads data were obtained from previous project AB2-104.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AC2-142

3. Loadflow and Dynamics Case Setup

The dynamics simulation analysis was carried out using PSS/E Version 33.7.0.

The load flow scenarios and fault cases for this study are based on PJM's Regional Transmission Planning Process³ and discussions with PJM.

The selected load flow scenario is the RTEP 2020 Light Load case with the following modifications:

- a) Addition of all applicable queue projects prior to AC1-003.
- b) Addition of AC1-003 queue project.
- c) Removal of withdrawn and subsequent queue projects in the vicinity of AC1-003.
- d) Dispatch of units in the PJM system in order to maintain slack generators within limits.
- e) Merchant transmission projects X3-028 and S57/S58 set online and at maximum power import into PJM.

The AC1-003 initial conditions are listed in Table 2, indicating maximum power output.

Table 2: AC1-003 machine initial conditions

Scenario	Bus	Name	Unit	PGEN (MW)	QGEN (MVar)	ETERM (pu)	POI Voltage (pu)
1	925355	AC1-003 CT1	1	278.14	-8.20	0.99	1.001
	925354	AC1-003 CT2	1	278.14	-8.20	0.99	
	925353	AC1-003 ST1	1	291.72	-6.57	0.99	

Generation within the vicinity of AC1-003 was dispatched online at maximum output (P_{MAX}). The dispatch of generation in the vicinity of AC1-003 is given in Attachment 5.

³ Manual 14B: PJM Region Transmission Planning Process, Rev 33, May 5 2016, Attachment G : PJM Stability, Short Circuit, and Special RTEP Practices and Procedures.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AC2-142

4. Fault Cases

Table 3 to Table 8 list the contingencies that were studied, with representative worst case total clearing times provided by PJM. Each contingency was studied over a 20 second simulation time interval.

The studied contingencies include:

- g) Steady state operation (20 second simulation);
- h) Three-phase faults with normal clearing time;
- i) Single-phase faults with stuck breaker;
- j) Single-phase faults placed at 80% of the line with delayed (Zone 2) clearing at line end remote from fault due to primary communications/relay failure;
- k) Single-phase faults resulting in tower failure;
- l) Three-phase faults with unsuccessful High Speed Reclosing (HSR).

No relevant Bus contingencies were identified.

Buses at which the faults listed above will be applied are:

- AC1-003 POI 345 kV
- Tidd 345 kV
- Wylie Ridge 345 kV
- Y2-050 POI 345 kV
- Toronto 345 kV

The three phase faults with normal clearing time were performed under network intact conditions.

Additional delayed (Zone 2) clearing at remote end faults were applied on lines from Holloway 345 kV, West Bellaire 345 kV, Collier 345 kV, and Canton Central 345 kV, and Sammis 345 kV towards the queue project.

A complete list of the contingencies that were studied and the one-line diagram of the APS network in the vicinity of AC1-003 is included in Attachment 2.

Clearing times listed in Tables 3 to 7 are as per revision 19 of “2016 Revised Clearing times for each PJM company” spreadsheet.

The positive sequence fault impedances for single line to ground faults were derived from the 2020 Light Load case.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AC2-142

5. Evaluation Criteria

This study is focused on AC1-003, along with the rest of the PJM system, maintaining synchronism and having all states return to an acceptable new condition following the disturbance. The recovery criteria applicable to this study are as per PJM's Regional Transmission Planning Process:

- a) The system with AC1-003 included should be transiently stable with post-contingency oscillations positively damped with a margin of at least 3%.
- b) The AC1-003 generator is able to ride through the faults (except for faults where protective action trips the generator(s)).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of the fault.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AC2-142

6. Summary of Results

For the fault contingencies tested on the 2020 Light Load case:

- a) Post-contingency oscillations were insufficiently damped for all contingencies. Pre- and post-AC1-003 damping analysis carried out on the CDG3 unit (bus 243185) showed that the oscillations remained, therefore this is not attributable to the addition of AC1-003.
- b) The AC1-003 generator was able to ride through all faults (except for faults where protective action trips a generator(s)).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of the fault.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AC2-142

7. Recommendations and Mitigations

No mitigations required

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AC2-142

Table 3: Steady State Operation

Fault ID	Duration	Result No mitigation
SS.01	Steady state 20 sec	Stable

Table 4: Three-Phase Faults with Normal Clearing

Fault ID	Fault description	Clearing Time (Cycles)	Result No mitigation
3N.01	Fault at AC1-003 POI 345 kV on AC1-003 circuit (trips AC1-003).	3	Insufficient Damping ¹ (Trips AC1-003)
3N.02	Fault at AC1-003 POI 345 kV on Wylie Ridge circuit.	3	Insufficient Damping ¹
3N.03	Fault at AC1-003 POI 345 kV on Tidd circuit.	3	Insufficient Damping ¹
3N.04	Fault at Tidd 345 kV on AC1-003 POI circuit.	4.5	Insufficient Damping ¹
3N.05	Fault at Tidd 345 kV on Y2-050 POI circuit.	4.5	Insufficient Damping ¹
3N.06	Fault at Tidd 345 kV on Holloway circuit.	4.5	Insufficient Damping ¹
3N.07	Fault at Tidd 345 kV on West Bellaire circuit.	4.5	Insufficient Damping ¹
3N.08	Fault at Tidd 345 kV on Collier circuit.	4.5	Insufficient Damping ¹
3N.09	Fault at Tidd 345 kV on Tidd Unit 2.	4.5	Insufficient Damping ¹
3N.10	Fault at Tidd 345 kV on Tidd Unit 3.	4.5	Insufficient Damping ¹
3N.11	Fault at Tidd 345 kV on Tidd 345/138 kV Transformer C.	4.5	Insufficient Damping ¹

¹ Insufficient damping was observed at AC1-003. Damping calculations were performed pre- and post- AC1-003 on CDG3 (bus 243185) which showed that the oscillations remained, and thus were not attributable to the addition of AC1-003.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AC2-142

Table 4 (Continued) : Three-Phase Faults with Normal Clearing

Fault ID	Fault description	Clearing Time (Cycles)	Result No mitigation ¹
3N.12	Fault at Wylie Ridge 345 kV on AC1-003 POI circuit.	3	Insufficient Damping ¹
3N.13	Fault at Wylie Ridge 345 kV on Toronto circuit.	3	Insufficient Damping ¹
3N.14	Fault at Wylie Ridge 345 kV on Wylie Ridge 500/345 kV Transformer 5 and Wylie Ridge 500/345 kV Transformer 6.	3	Insufficient Damping ¹
3N.15	Fault at Wylie Ridge 345 kV on Wylie Ridge 500/345 kV Transformer 7 and Wylie Ridge 500/345 kV Transformer 8.	3	Insufficient Damping ¹
3N.16	Fault at Wylie Ridge 345 kV on Wylie Ridge 345/138 kV Transformer 1 and Wylie Ridge 345/138 kV Transformer 2.	3	Insufficient Damping ¹
3N.17	Fault at Wylie Ridge 345 kV on Wylie Ridge 345/138 kV Transformer 3.	3	Insufficient Damping ¹
3N.18	Fault at Y2-050 POI 345 kV on Tidd circuit.	4.5	Insufficient Damping ¹
3N.19	Fault at Y2-050 POI 345 kV on Canton Central circuit.	4.5	Insufficient Damping ¹
3N.20	Fault at Y2-050 POI 345 kV on Y2-050 circuit (trips Y2-050 generation).	4.5	Insufficient Damping ¹
3N.21	Fault at Toronto 345 kV on Wylie Ridge circuit.	3	Insufficient Damping ¹
3N.22	Fault at Toronto 345 kV on Sammis circuit.	3	Insufficient Damping ¹
3N.23	Fault at Toronto 345 kV on Toronto 345/138 kV Transformer 1.	3	Insufficient Damping ¹
3N.24	Fault at Toronto 345 kV on Toronto 345/138 kV Transformer 2.	3	Insufficient Damping ¹

¹ Insufficient damping was observed at AC1-003. Damping calculations were performed pre- and post- AC1-003 on CDG3 (bus 243185) which showed that the oscillations remained, and thus were not attributable to the addition of AC1-003.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AC2-142

Table 5: Single-Phase Faults with Stuck Breaker

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	Result No mitigation
1B.01	Fault at AC1-003 POI 345 kV on AC1-003 circuit. Breaker stuck to Tidd circuit. Fault cleared with loss of Tidd circuit (trips AC1-003).	3 / 12	Insufficient Damping ¹ (Trips AC1-003)
1B.02	Fault at AC1-003 POI 345 kV on AC1-003 circuit. Breaker stuck to Wylie Ridge circuit. Fault cleared with loss of Wylie Ridge circuit (trips AC1-003).	3 / 12	Insufficient Damping ¹ (Trips AC1-003)
1B.03	Fault at AC1-003 POI 345 kV on Wylie Ridge circuit. Breaker stuck to AC1-003 circuit. Fault cleared with loss of AC1-003 circuit (trips AC1-003).	3 / 12	Insufficient Damping ¹ (Trips AC1-003)
1B.04	Fault at AC1-003 POI 345 kV on Wylie Ridge circuit. Breaker stuck to Tidd circuit. Fault cleared with loss of Tidd circuit (trips AC1-003).	3 / 12	Insufficient Damping ¹ (Trips AC1-003)
1B.05	Fault at AC1-003 POI 345 kV on Tidd circuit. Breaker stuck to AC1-003 circuit. Fault cleared with loss of AC1-003 circuit (trips AC1-003).	3 / 12	Insufficient Damping ¹ (Trips AC1-003)
1B.06	Fault at AC1-003 POI 345 kV on Tidd circuit. Breaker stuck to Wylie Ridge circuit. Fault cleared with loss of Wylie Ridge circuit (trips AC1-003).	3 / 12	Insufficient Damping ¹ (Trips AC1-003)
1B.07	Fault at Tidd 345 kV on AC1-003 POI circuit. Breaker CC stuck. Fault cleared with loss of Collier circuit.	4.5 / 15	Insufficient Damping ¹
1B.08	Fault at Tidd 345 kV on AC1-003 POI circuit. Breaker CC2 stuck. Fault cleared with loss of Tidd 345/138 kV Transformer C.	4.5 / 15	Insufficient Damping ¹
1B.09	Fault at Tidd 345 kV on Y2-050 POI circuit. Breaker AA stuck. Fault cleared with loss of Holloway circuit.	4.5 / 15	Insufficient Damping ¹
1B.10	Fault at Tidd 345 kV on Holloway circuit. Breaker AA stuck. Fault cleared with loss of Y2-050 POI circuit.	4.5 / 15	Insufficient Damping ¹
1B.11	Fault at Tidd 345 kV on Holloway circuit. Breaker AA2 stuck. Fault cleared with loss of Tidd 345/138 kV Transformer C.	4.5 / 15	Insufficient Damping ¹
1B.12	Fault at Tidd 345 kV on West Bellaire circuit. Breaker BB stuck. Fault cleared with loss of Tidd Unit 2.	4.5 / 15	Insufficient Damping ¹
1B.13	Fault at Tidd 345 kV on West Bellaire circuit. Breaker BB2 stuck. Fault cleared with loss of Tidd 345/138 kV Transformer C.	4.5 / 15	Insufficient Damping ¹
1B.14	Fault at Tidd 345 kV on Collier circuit. Breaker CC stuck. Fault cleared with loss of AC1-003 POI circuit.	4.5 / 15	Insufficient Damping ¹
1B.15	Fault at Tidd 345 kV on Tidd Unit 2. Breaker BB stuck. Fault cleared with loss of West Bellaire circuit.	4.5 / 15	Insufficient Damping ¹
1B.16	Fault at Tidd 345 kV on Tidd Unit 3. Breaker FF2 stuck. Fault cleared with loss of Tidd 345/138 kV Transformer C.	4.5 / 15	Insufficient Damping ¹
1B.17	Fault at Tidd 345 kV on Tidd 345/138 kV Transformer C. Breaker AA2. Fault cleared with loss of Holloway circuit.	4.5 / 15	Insufficient Damping ¹
1B.18	Fault at Tidd 345 kV on Tidd 345/138 kV Transformer C. Breaker BB2. Fault cleared with loss of West Bellaire circuit.	4.5 / 15	Insufficient Damping ¹

¹ Insufficient damping was observed at AC1-003. Damping calculations were performed pre- and post- AC1-003 on CDG3 (bus 243185) which showed that the oscillations remained, and thus were not attributable to the addition of AC1-003.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AC2-142

Table 5 (Continued): Single-Phase Faults with Stuck Breaker

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	Result No mitigation
1B.19	Fault at Tidd 345 kV on Tidd 345/138 kV Transformer C. Breaker CC2. Fault cleared with loss of AC1-003 POI circuit.	4.5 / 15	Insufficient Damping ¹
1B.20	Fault at Tidd 345 kV on Tidd 345/138 kV Transformer C. Breaker FF2. Fault cleared with loss Tidd Unit 3.	4.5 / 15	Insufficient Damping ¹
1B.21	Fault at Wylie Ridge 345 kV on AC1-003 POI circuit. Breaker WK5 stuck to Wylie Ridge 500/345 kV Transformer 7 and Wylie Ridge 500/345 kV Transformer 8. Fault cleared with loss of Wylie Ridge 500/345 kV Transformer 7 and Wylie Ridge 500/345 kV Transformer 8.	3 / 12	Insufficient Damping ¹
1B.22	Fault at Wylie Ridge 345 kV on Toronto circuit. Breaker WK2 stuck. Fault cleared with loss of Wylie Ridge 345/138 kV Transformer 1 and Wylie Ridge 345/138 kV Transformer 2.	3 / 12	Insufficient Damping ¹
1B.23	Fault at Wylie Ridge 345 kV on Toronto circuit. Breaker WK3 stuck. Fault cleared with loss of Wylie Ridge 345/138 kV Transformer 3.	3 / 12	Insufficient Damping ¹
1B.24	Fault at Wylie Ridge 345 kV on Wylie Ridge 500/345 kV Transformer 5 and Wylie Ridge 500/345 kV Transformer 6. Breaker WK4. Fault cleared with loss of AC1-003 POI circuit.	3 / 12	Insufficient Damping ¹
1B.25	Fault at Wylie Ridge 345 kV on Wylie Ridge 500/345 kV Transformer 5 and Wylie Ridge 500/345 kV Transformer 6. Breaker WK1. Fault cleared with loss of Tidd 345/138 kV Transformer 1 and Tidd 345/138 kV Transformer 2.	3 / 12	Insufficient Damping ¹
1B.26	Fault at Wylie Ridge 345 kV on Wylie Ridge 500/345 kV Transformer 7 and Wylie Ridge 500/345 kV Transformer 8. Breaker WK5. Fault cleared with loss of AC1-003 POI circuit.	3 / 12	Insufficient Damping ¹
1B.27	Fault at Wylie Ridge 345 kV on Wylie Ridge 500/345 kV Transformer 7 and Wylie Ridge 500/345 kV Transformer 8. Breaker WK6. Fault cleared with loss of Tidd 345/138 kV Transformer 3.	3 / 12	Insufficient Damping ¹

¹ Insufficient damping was observed at AC1-003. Damping calculations were performed pre- and post- AC1-003 on CDG3 (bus 243185) which showed that the oscillations remained, and thus were not attributable to the addition of AC1-003.

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	Result No mitigation
1B.28	Fault at Wylie Ridge 345 kV on Wylie Ridge 345/138 kV Transformer 1 and Wylie Ridge 345/138 kV Transformer 2. Breaker WK1 stuck. Fault cleared with loss of Wylie Ridge 500/345 kV Transformer 5 and Wylie Ridge 500/345 kV Transformer 6.	3 / 12	Insufficient Damping ¹
1B.29	Fault at Wylie Ridge 345 kV on Wylie Ridge 345/138 kV Transformer 1 and Wylie Ridge 345/138 kV Transformer 2. Breaker WK2. Fault cleared with loss of Toronto circuit.	3 / 12	Insufficient Damping ¹
1B.30	Fault at Wylie Ridge 345 kV on Wylie Ridge 345/138 kV Transformer 3. Breaker WK6 stuck. Fault cleared with loss of Wylie Ridge 500/345 kV Transformer 7 and Wylie Ridge 500/345 kV Transformer 8.	3 / 12	Insufficient Damping ¹
1B.31	Fault at Wylie Ridge 345 kV on Wylie Ridge 345/138 kV Transformer 3. Breaker WK3 stuck. Fault cleared with loss of Toronto circuit.	3 / 12	Insufficient Damping ¹
1B.32	Fault at Y2-050 POI 345 kV on Tidd circuit. Breaker stuck to Canton Central circuit. Fault cleared with loss of Canton Central circuit (trips Y2-050).	4.5 / 15	Insufficient Damping ¹
1B.33	Fault at Y2-050 POI 345 kV on Tidd circuit. Breaker stuck to Y2-050 circuit. Fault cleared with loss of Y2-050 circuit (trips Y2-050).	4.5 / 15	Insufficient Damping ¹
1B.34	Fault at Y2-050 POI 345 kV on Canton Central circuit. Breaker stuck to Tidd circuit. Fault cleared with loss of Tidd circuit (trips Y2-050).	4.5 / 15	Insufficient Damping ¹
1B.35	Fault at Y2-050 POI 345 kV on Canton Central circuit. Breaker stuck to Y2-050 circuit. Fault cleared with loss of Y2-050 circuit (trips Y2-050).	4.5 / 15	Insufficient Damping ¹
1B.36	Fault at Y2-050 POI 345 kV on Y2-050 circuit. Breaker stuck to Tidd circuit. Fault cleared with loss of Tidd circuit (trips Y2-050).	4.5 / 15	Insufficient Damping ¹

¹ Insufficient damping was observed at AC1-003. Damping calculations were performed pre- and post- AC1-003 on CDG3 (bus 243185) which showed that the oscillations remained, and thus were not attributable to the addition of AC1-003.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AC2-142

Table 5 (Continued): Single-Phase Faults with Stuck Breaker

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	Result No mitigation
1B.37	Fault at Y2-050 POI 345 kV on Y2-050 circuit. Breaker stuck to Canton Central circuit. Fault cleared with loss of Canton Central circuit (trips Y2-050).	4.5 / 15	Insufficient Damping ¹
1B.38	Fault at Toronto 345 kV on Wylie Ridge circuit. Breaker 5. Fault cleared with loss of Toronto 345/138 kV Transformer 1.	3 / 12	Insufficient Damping ¹
1B.39	Fault at Toronto 345 kV on Wylie Ridge circuit. Breaker 6. Fault cleared with loss of Toronto 345/138 kV Transformer 2.	3 / 12	Insufficient Damping ¹
1B.40	Fault at Toronto 345 kV on Sammis circuit. Breaker 4 stuck. Fault cleared with loss of Toronto 345/138 kV Transformer 2.	3 / 12	Insufficient Damping ¹
1B.41	Fault at Toronto 345 kV on Sammis circuit. Breaker 3 stuck. Fault cleared with loss of Toronto 345/138 kV Transformer 1.	3 / 12	Insufficient Damping ¹
1B.42	Fault at Toronto 345 kV on Toronto 345/138 kV Transformer 1. Breaker 3 stuck. Fault cleared with loss of Sammis circuit.	3 / 12	Insufficient Damping ¹
1B.43	Fault at Toronto 345 kV on Toronto 345/138 kV Transformer 1. Breaker 5 stuck. Fault cleared with loss of Wylie Ridge circuit.	3 / 12	Insufficient Damping ¹
1B.44	Fault at Toronto 345 kV on Toronto 345/138 kV Transformer 2. Breaker 4 stuck. Fault cleared with loss of Sammis circuit.	3 / 12	Insufficient Damping ¹
1B.45	Fault at Toronto 345 kV on Toronto 345/138 kV Transformer 2. Breaker 6 stuck. Fault cleared with loss of Wylie Ridge circuit.	3 / 12	Insufficient Damping ¹

¹ Insufficient damping was observed at AC1-003. Damping calculations were performed pre- and post- AC1-003 on CDG3 (bus 243185) which showed that the oscillations remained, and thus were not attributable to the addition of AC1-003.

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AC2-142

Table 6: Single-Phase Faults with Delayed (Zone2) Clearing at line end closest to AC1-003 POI

Fault ID	Fault description	Clearing Time normal & delayed (Cycles)	Result No mitigation
1D.01	Fault at 80% of 345 kV line from AC1-003 POI 345 kV to Wylie Ridge circuit. Delayed clearing at AC1-003 POI 345 kV.	3 / 21	Insufficient Damping ¹
1D.02	Fault at 80% of 345 kV line from AC1-003 POI 345 kV to Tidd circuit. Delayed clearing at AC1-003 POI 345 kV.	4.5 / 21	Insufficient Damping ¹
1D.03	Fault at 80% of 345 kV line from Tidd 345 kV to Y2-050 POI circuit. Delayed clearing at Tidd 345 kV.	4.5 / 60	Insufficient Damping ¹
1D.04	Fault at 80% of 345 kV line from Tidd 345 kV to Holloway circuit. Delayed clearing at Tidd 345 kV.	4.5 / 60	Insufficient Damping ¹
1D.05	Fault at 80% of 345 kV line from Tidd 345 kV to West Bellaire circuit. Delayed clearing at Tidd 345 kV.	4.5 / 60	Insufficient Damping ¹
1D.06	Fault at 80% of 345 kV line from Tidd 345 kV to Collier circuit. Delayed clearing at Tidd 345 kV.	4.5 / 60	Insufficient Damping ¹
1D.07	Fault at 80% of 345 kV line from Wylie Ridge 345 kV to Toronto circuit. Delayed clearing at Wylie Ridge 345 kV.	3 / 21	Insufficient Damping ¹
1D.08	Fault at 80% of 345 kV line from Y2-050 POI 345 kV to Canton Central circuit. Delayed clearing at Y2-050 POI 345 kV.	4.5 / 60	Insufficient Damping ¹
1D.09	Fault at 80% of 345 kV line from Toronto 345 kV to Sammis circuit. Delayed clearing at Toronto 345 kV.	3 / 21	Insufficient Damping ¹

¹ Insufficient damping was observed at AC1-003. Damping calculations were performed pre- and post- AC1-003 on CDG3 (bus 243185) which showed that the oscillations remained, and thus were not attributable to the addition of AC1-003.

Table 7: Single-Phase Faults Resulting in Tower Failure

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	Result No mitigation
1T.01	Fault at AC1-003 POI 345 kV on Wylie Ridge circuit resulting in tower failure. Fault cleared with additional loss of Tidd – Collier circuit. CONTINGENCY 'C5-MP-138-054_A' /* TIDD-WR-TIDD-COL	3	Insufficient Damping ¹
1T.02	Fault at AC1-003 POI 345 kV on Tidd circuit resulting in tower failure. Fault cleared with additional loss of Tidd – Collier circuit. CONTINGENCY 'C5-MP-138-054_B' /* TIDD-WR-TIDD-COL	3	Insufficient Damping ¹
1T.03	Fault at Tidd 345 kV on Holloway circuit resulting in tower failure. Fault cleared with additional loss of Tidd - West Bellaire circuit. CONTINGENCY '8945'	4.5	Insufficient Damping ¹
1T.04	Fault at Tidd 345 kV on Holloway circuit resulting in tower failure. Fault cleared with additional loss of Tidd – West Bellaire circuit. CONTINGENCY '8955'	4.5	Insufficient Damping ¹

Appendix 3 (Continued)

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AC2-142

Table 7: Three-Phase Faults with Unsuccessful High Speed Reclosing

Fault ID	Fault description	Clearing Time normal & reclose (Cycles)	Result No mitigation
3R.01	Fault at Tidd 345 kV on AC1-003 POI circuit. - Fault cleared after 4.5 cycles with loss of AC1-003 POI circuit. - High speed reclosers CC and CC2 close after 21 cycles (25.5 cycles total) reconnecting AB2-104 POI circuit. - Reclose unsuccessful, fault cleared after 4.5 cycles (30 cycles total) by opening AC1-003 POI circuit.	4.5 / 21	Insufficient Damping ¹
3R.02	Fault at Tidd 345 kV on Y2-050 POI circuit. - Fault cleared after 4.5 cycles with loss of Y2-050 POI circuit - High speed reclosers AA and AA1 close after 21 cycles (25.5 cycles total) reconnecting Y2-050 POI circuit. - Reclose unsuccessful, fault cleared after 4.5 cycles (30 cycles total) by opening Y2-050 POI circuit.	4.5 / 21	Insufficient Damping ¹
3R.03	Fault at Tidd 345 kV on Collier circuit. - Fault cleared after 4.5 cycles with loss of Collier circuit - High speed reclosers CC and CC1 close after 21 cycles (25.5 cycles total) reconnecting Collier circuit. - Reclose unsuccessful, fault cleared after 4 cycles (30 cycles total) by opening Collier circuit.	4.5 / 21	Insufficient Damping ¹
3R.04	Fault at Tidd 345 kV on Holloway circuit. - Fault cleared after 4.5 cycles with loss of Holloway circuit - High speed reclosers AA and AA2 close after 21 cycles (25.5 cycles total) reconnecting Holloway circuit. - Reclose unsuccessful, fault cleared after 4.5 cycles (30 cycles total) by opening Holloway circuit.	4.5 / 21	Insufficient Damping ¹
3R.05	Fault at Tidd 345 kV on West Bellaire circuit. - Fault cleared after 4.5 cycles with loss of West Bellaire circuit - High speed reclosers BB and BB2 close after 21 cycles (25.5 cycles total) reconnecting West Bellaire circuit. - Reclose unsuccessful, fault cleared after 4.5 cycles (30 cycles total) by opening West Bellaire circuit.	4.5 / 21	Insufficient Damping ¹
3R.06	Fault at Wylie Ridge 345 kV on AC1-003 POI circuit. - Fault cleared after 3 cycles with loss of AC1-003 POI circuit. - High speed reclosers WK4 and WK5 close after 12 cycles (15 cycles total) reconnecting AC1-003 POI circuit². - Reclose unsuccessful, fault cleared after 3 cycles (18 cycles total) by opening AC1-003 POI circuit.	3 / 12	Insufficient Damping ¹
3R.07	Fault at Wylie Ridge 500 kV on AB1-069 POI circuit. - Fault cleared after 3 cycles with loss of AB1-069 POI circuit. - High speed reclosers WL1 close after 2 cycles (5 cycles total) reconnecting AB1-069 POI circuit². - Reclose unsuccessful, fault cleared after 3 cycles (8 cycles total) with loss of AB1-069 POI circuit. - High speed reclosers WL2 close after 2 cycles (10 cycles total) reconnecting AB1-069 POI circuit². - Reclose unsuccessful, fault cleared after 3 cycles (13 cycles total) by opening AB1-069 POI circuit.	3 / 2	Insufficient Damping ¹

¹ Insufficient damping was observed at AC1-003. Damping calculations were performed pre- and post- AC1-003 on CDG3 (bus 243185) which showed that the oscillations remained, and thus were not attributable to the addition of AC1-003.

² It was assumed the remote end also closes with high speed reclosing

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